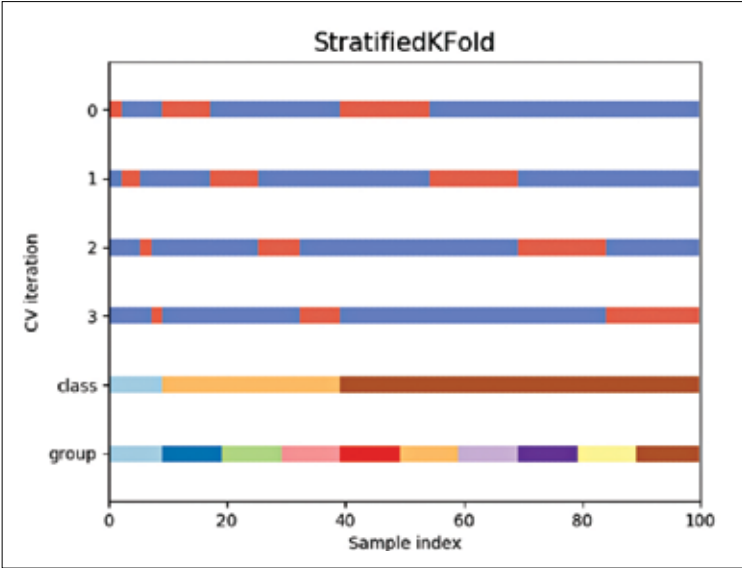


Above validation technique is non-exhaustive as it does not compute all ways of splitting the original sample. We have to decide how many subsets need to be made.



Stratified K-Fold cross validation

Leave-P-Out Cross Validation

In this approach, all of the data except p records are used for training and p records are used for testing. This process is repeated for all combinations possible. The advantage is that entire data is used for training and testing just like k-fold cross validation. The error rate of the model is the average of the error rate of each iteration, to give overall effectiveness.

This method is infeasible for higher values of p as all possible combinations have to be validated and finding all possible cases take exponential time complexity.

A special case is of $p = 1$ is known as leave-1-out-cross validation. It is preferred over $p > 1$ because this process is only repeated for N times if there are N records. This reduces the unnecessary exponential intensive processing while keeping the advantages of using the entire dataset for training and testing.